This lecture provides a comprehensive overview of several statistical tests, including Z-tests, Chi-square tests, and ANOVA (Analysis of Variance). The lecture aims to clarify when and how to use each test, emphasizing the underlying principles and calculations involved.

Key Concepts and Topics:

1. Review of T-tests:
    - Briefly revisits the T-test (Student's t-distribution), assuming prior knowledge from a previous class.
    - Acknowledges that some students may need a revision of T-tests and encourages them to review materials.
2. Introduction to Z-tests:
    - Explains that Z-tests are used for large samples or when the population standard deviation is known.
    - Presents a detailed example of a Z-test, including:
        - Formulating a null hypothesis (e.g., students score more than 70 on average).
        - Collecting sample data.
        - Calculating the Z-score using the formula: Z = (sample mean - population mean) / (population standard deviation / sqrt(sample size)).
        - Using a Z-table to find the critical Z-value.
        - Making a decision to accept or reject the null hypothesis based on the comparison of the calculated Z-score and the critical Z-value.
        - Discusses the importance of the level of confidence (alpha) in hypothesis testing and how it affects the critical Z-value and the decision to accept or reject the null hypothesis.
3. Chi-Square Tests:
    - Introduces Chi-square tests for categorical data.
    - Presents an example involving student preferences for mathematics based on gender (boys vs. girls liking or disliking math).
    - Explains how to create a contingency table to summarize the data.
    - Formulates null and alternate hypotheses (e.g., gender and liking math are independent vs. there is a relationship).
    - Explains the Chi-square formula: $\chi^2 = \Sigma((\text{observed} - \text{expected})^2 / \text{expected})$.
    - Details how to calculate expected values using the formula: (row total * column total) / grand total.
    - Calculates the Chi-square statistic and compares it to a critical value from a Chi-square table.
    - Determines the degrees of freedom (df = (rows - 1) * (columns - 1)).
    - Makes a decision to accept or reject the null hypothesis based on the comparison of the calculated Chi-square value and the critical Chi-square value.
4. ANOVA (Analysis of Variance):
    - Explains that ANOVA is used when comparing more than two categories or groups.
    - Presents an example involving three different teaching methods (A, B, C) and their impact on student scores.

- Formulates null and alternate hypotheses (e.g., all teaching methods have the same impact vs. at least one method is different).
- Calculates group means and the grand mean.
- Explains the core philosophy of ANOVA: analyzing variance between groups and within groups.
- Details the calculation of the sum of squares within (SSW) and the sum of squares between (SSB).
- Calculates degrees of freedom between (dfb) and degrees of freedom within (dfw).
- Calculates the mean square between (MSB) and the mean square within (MSW).
- Calculates the F-statistic: F = MSB / MSW.
- Compares the calculated F-statistic to a critical F-value from an F-table.
- Makes a decision to accept or reject the null hypothesis based on the comparison of the calculated F-statistic and the critical F-value.

Main Learning Objectives and Takeaways:

- Understand the appropriate use cases for Z-tests, Chi-square tests, and ANOVA.
- Be able to formulate null and alternate hypotheses.
- Know how to calculate test statistics (Z, Chi-square, F) and interpret their values.
- Be able to use statistical tables (Z, Chi-square, F) to find critical values.
- Understand the importance of the level of confidence (alpha) in hypothesis testing.
- Be able to make informed decisions to accept or reject null hypotheses based on test results.
- Understand the underlying principles and assumptions of each statistical test.
- Recognize the importance of these statistical concepts in data analysis and machine learning.